Milestone II Submission

Team Name

Ah Giraffe

Proposed Level of Achievement

Apollo

Motivation

Having used many reminder applications to manage our daily tasks, we found out that the most challenging part of such application is to encourage users to presevere. Oftentimes, users will forget or lack the motivation to tick the boxes after a period of time. How can we encourage users to complete their goals, either short-term or long-term? As both of us are animal lovers, we thought that if such elements can be integrated into the reminder application, we would be more motivated to complete the goals we set. Also, we realised that by including a section on our application to track long-term goals, it could be very useful in helping users cultivate a healthy lifestyle, such as exercising 30 hours per month.

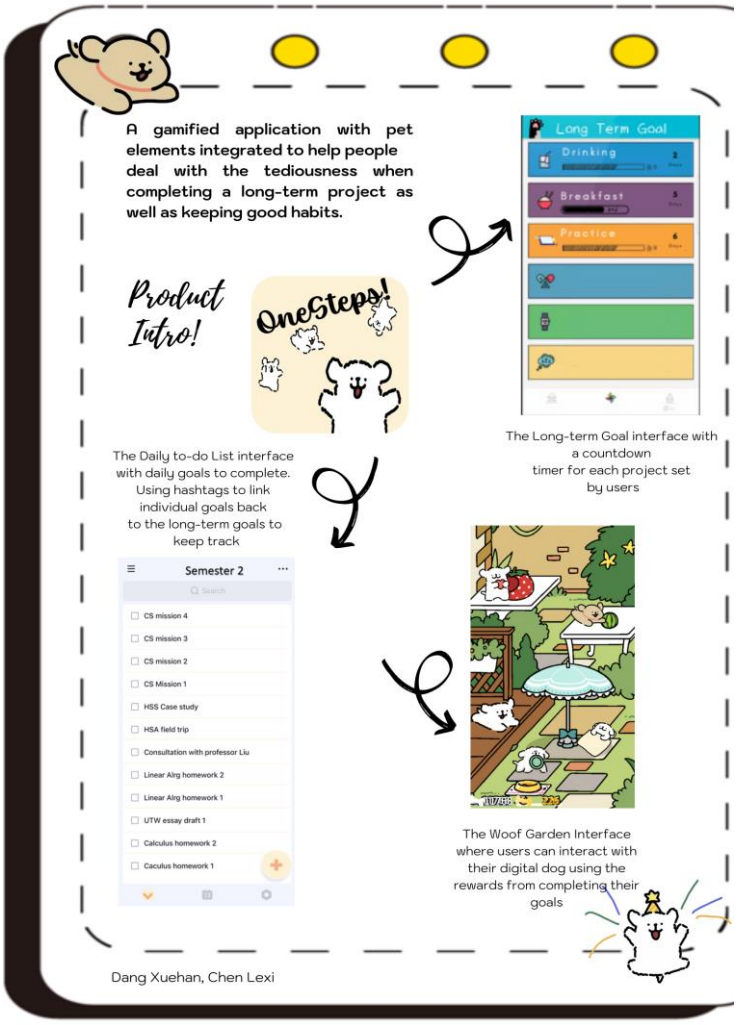
Vision

Our proposed application is named Onesteps. It will be a reminder mobile application with pet elements integrated. It will also include a gamified component which gives users more incentives to use the application. It will have **three interfaces**, and **pet characters** which users can obtain using the points accumulated from completing their tasks. We are also looking to add in **interactive features** such as feeding to make our application more engaging. Our vision is to help users better manage their tasks. Every step counts no matter big or small, and you just have to make one step per day!

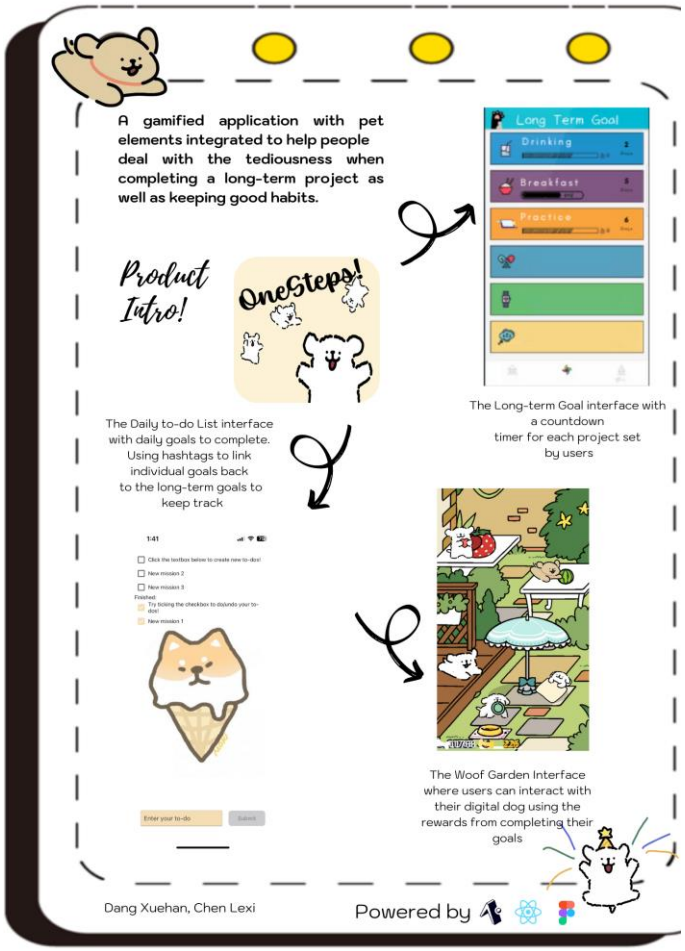
App Logo

Onesteps App Logo

Lift-off Poster



Milestone 1 Poster



Milestone 1 poster

ABOUT

Onesteps is an all in one reminder application aiming to help users keep track of their daily tasks and long-term goals. Gamified pet element is integrated to make it more appealing.

Reward System is enrolled in *Onestep*.
In *Onestep*, users can be rewarded when completing tasks .
-1 bone is rewarded upon completing each to-do, 10 is rewarded upon completing each long -term goal.
-Bones can be used in Woof garden to interact with the pet.
-Each interaction costs 1 bone.

FEATURES

- Local Storage
- To-do List
- Long Term Goal interface
- Reward System
- Interactive Woof Garden Interface

TECH STACK

To-do List

1. Keep track of daily to-dos.
Users can tick the box when they complete a task they created.
A confirmation window will appear when ticking the boxes to avoid false touch.
Finished task cannot be unchecked.

Long-term Goal

2. Set up long-term goals.
This interface alllow user to set a long term goal such as running 30 km in a month.
Each time they finish part of the goal, e.g 5km, they can key in the amount and click the 'paw' icon to clock their work.
A progress bar will show the percentage of how much they have completed.

Woof Garden

3. Woof garden allows user to use thier reward to interact with the pet.
Functions currently implemented include Bathing, Feeding and Walking.
A 'Happiness Level' (HL) bar is located at the top-right corner.
A certain level of interaction is needed to maintain the HL of the pet.
Each interaction will add 5% of happiness.
20% happiness will automatically drop every day.

Presented by Ah Giraffe

Milstone 2 poster

User Stories

1.As a student who has many long-term projects to complete, I want to be able to have an overview of my plan as well as specific daily tasks to complete.

2.As a young person, I prefer applications that are anchored with gamification along with interesting features rather than plain interfaces and basic functions.

3.As a dog lover, being able to be awarded to keep a digital pet motivates me to complete certain tasks.

4.As a person who lives in this modern society, I want to have an application that facilitates me to develop good and healthy habits over time to improve my living quality.

Features

1. **Long-term Goal**

   This interface allows users to set a timeline for their long-term goals. For example, running 30km in 1 month. In the main page, there will be a progress bar to show how many percent of the goal have been completed. When clicked on the long-term goal, users will be directed to a in-progress page where they can tick the box for completing a smaller goal. For example, running 3km each day. Succesfully completing each goal will reward users points to be used in the Woof Garden which will be elloborated on later.

2. **Daily to-do List**

   This interface allows users to plan our their agenda for the day. They can tick the boxes for tasks they have achieved. Similar to above, succesfully completing each task will reward users points to be used in the Woof Garden

3. **Woof Garden**

   This interface is where the gamified and interactive features come in. There will be different pet characters where users can obtain using the points earned from the above interfaces.User can also interact with the characters and feed them regurlarly to maintain the Happiness Level (HL). There will be a minimum HL to be maintained, or the character will be disabled until actions are done to fulfill the minimum HL. This could encourage users to complete their goals on time.

Proof of Concept

Preview release channel "default"

Open the camera app on your device and scan the code below

https://expo.dev/@xuehann/orbital-app?serviceType=classic&distribution=expo-go

User Instruction:

1. Download Expo Go on your IOS or Android device
2. Scan the above QR code
3. Or click on the above link
4. PS: Due to high quality of the animation used, please use mobile data to evaluate this work unless the wifi connection is very good! Thanks!

If the QR code does not work, please try the following:

1. You can follow yadunut's ReactNative mission control slides for detailed instructions in order to meet all pre-requisites:https://yadunut.github.io/orbital-react-native-2023/slides/setup.html#5
2. Create a directory to house our code.
3. Clone the repository from this ssh key: git@github.com:concealexi/Ah-Giraffe.git
4. Once cloned, change to the "orbital-app" directory. Type the below in your terminal

Or skip this step and open the folder using VSCode directly, jump to step 6

```
cd orbital-app
```

5. Then npm install and open up vscode:

```
npm install
```

```
code .
```

6. Open up your terminal/open the cloned folder in vscode and start expo go:

```
npx expo start
```

Or

```
npx expo start --tunnel
```

7. Scan the QR code produced and open Expo Go
8. Try OneStep!

(It will take a while to load the animation)

Work Log

Tasks	Lexi (hours)	Xuehan (hours)	Remarks
Planning, Video, Poster	5	5	1. Finalised ideas and concepts on gam 2. Make demonstration of interfaces 3. Make posters & Create video
Technical Knowledge Research	20	15	1. Watching workshops on React Nativ Expo 2. Learning the usage of Git and GitHu 3. Learning Uniapp and HBuilder 4. Picking up knowledge on Javascript 5. Figam specifications

Setting up Git	1	0	Setting up private Git Repository and ex student pack tools available
Woof Garden Design	2	0	Design the layout and features of the int
To-do list Design	2	0	Design the layout and features of the int
Long-term Goal Design	0	2	Design the layout and features of the int
To-do list implementation	6	0	Coding using Expo and related extensio
Meeting up	3	3	
Researching and resolving technical issues	3	3	Issues related to installation and depend
Learning Photoshop for basic digital art	1	1	
Character Design (In Progress) ● Pets ● Rewards ● Background	1	1	
After MS1(MS2) Redesigning of interface and use according to feedback	3	3	Refer to MS1 peer evaluation feedback
Figma design of all 3 interfaces (todolist done in MS1)	5	5	adjustments will be make accordingly in following steps if needed
Change the todolist interface according to feedback given+made during disccussion	5	0	

Create Long term goal interface	0	5*7	Further explore React native
Create Woof Garden Interface	5*7	0	Further explore React native, seek help outside adviser who knows react natice
Look for suitable image for Woof Garden	2	2	
Remove background of Image + Create animation	8	8	Learning Adoba After Effect, Adobe Ph Procreate
Added in local storage	5	5	
Connection of 3 interfaces	10	8	Seek help from outside adviser for sugg designing and coding
Reconsidering bugs and 'holes' in the design, user experience testing	5	5	consider what is the suitable range/numl bones to be awarded/deducted from all 3 modify according to user experience testing(audiences: friends and family me
New Poster design	3	5	Done together via Canva
New Pitch Video	1	2	

Milestone 1 peer evaluation:

1. have tested out the app and realise that as I add more tasks, the tasks will be out of the screen. maybe can try to make it scrollable so that users can view all their to-dos

2. from the readme, 'Successfully completing each goal will reward users points to be used in the Woof Garden'. If the user was to unclick the finished task and click it again, will points be given again? this could be a loophole for users to gain many points quickly.

3. however, the link does not work with iOS devices. i suspect it only works for android devices currently.

4. Yes a working component, with custom made art was shown (the todo-list). Even if the storage part was missing (since it can be added in very fast, I'll assume it is there) it is still a working component as majority of the work is already done.

5. I think a better idea than the 'Woof Garden' would be to focus on other features like Login and Storage. Things like this will allow for better professional training as these are tools most apps use globally. I think a better area to focus on to develop react-native apps would be to look at things like Firebase and AsyncStorage, and improve or building on the current features you have (the habit progress tracker thing, and the todo list).

6. Daily to-do List: Strongly Agree. Yes this is a really good way to show CRUD capabilities and is a very good tool to help people be more productive. It is already implemented with your art-style/theme (in the video) so this is done. Adding AsyncStorage (or another storage tool) will be very easy. The feature may seem a little simple because of how many times it is done, but it works with a clear benefit and is designed to the user's goal.

7. Woof Garden: Neutral. I don't think additional game-like features are going to help people be more productive. Also, how would game-like elements like this be implemented in react-native? I know progress bars are possible (one of the pictures you showed on your poster), but I am confused about the 'Woof Garden' part. Isn't that picture on your poster from another game? I think this form of reward system is too excessive and also outside the scope of CRUD. I might get people to just spend more time on their phones which is already a problem. But most of all I'm clueless as to how a system like this could be implemented into react-native.
8. I think the food for the pets is gained through completing the tasks (like the todo-list tasks). I would like to see more explanation of how this feature works (i.e. where the food for the pets comes from if there is any food, what the pets do, and how they motivate the user). The benefit seems unclear and the proposed design looks undefined (i.e. how will it be executed).

9. The app seems to be working but there was not any login/ authentication.

Also went i restarted the app, things that i added to the list was removed even though it was not checked

From Adviser:

Given no / minimal evidence (all teams expected in MS1, will definitely come in future milestones), Team will have to refer back to the drawing board on the software engineering practices (keep in mind this is expected), and not overlooking the expected requirements for Apollo achievement level. If team is unsure of jargons e.g., Activity flow, use cases, tech architecture, etc. they should search and clarify on those.

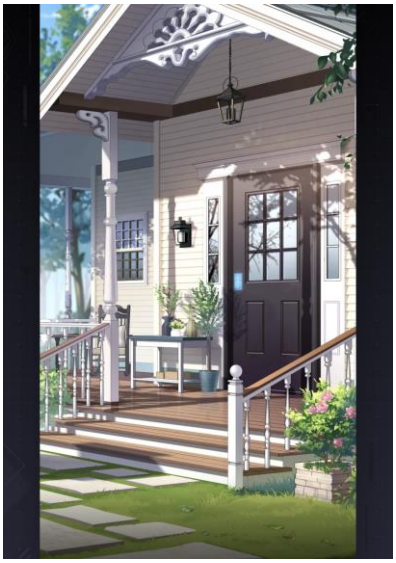
what to improve from peer/adviser evaluation after milestone 1：

1. make the todo list scrollable - react native flatlist
2. click/unclick task, points will not be repeatitively given!
3. add local storage so that when restarting the app, things added persists
4. work with both IOS and Android
5. how to create woof garden using react native???
6. no login/authentification -> turns out our team decide not to have it -> need to justify in our document

Milestone 2

Animation used (extracted from the myth of thermis - a Chinese Otome game)

JPG used from the myth of thermis - a Chinese Otome game



SVG used from ICONFONT

Links

Figma design link:

https://www.figma.com/file/s7oX3AuRMn2zTm29CHGxof/interface-design?type=design&node-id=9-3&t=gTodBphTRgp6qeRa-0

Poster design link:

https://www.canva.cn/design/DAFmy2kukMg/Az4tsBCAWuJITdUl9Du9mg/edit?utm_content=DAFmy2kukMg&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

Pitch Video link:

https://www.youtube.com/watch?v=otIsWdECgxo

Poster google drive link:

https://drive.google.com/file/d/1UTHyqkfnh9VXr0ZSSsVPq4ICzhGdJZ1n/view?usp=sharing

Things to note:

1. As we used gif animation, the file size is quite big, it works smoothly under mobile data, but tend to stuck under wifi, finding ways to solve this issue
2. Image quality pending improvement
3. Delete function is expected to be added in later for to-do
4. Added in local storage
5. To-do list can scroll down
6. JUSTIFICATION for lack of login feature: we currently believe that people seldom use reminder applications on multiple devices, hence we beleive that local storage would suffice

Notes for MS2:

1. In Long term goal interface, all daily tasks can only be submitted once in a day. This is to ensure that the Habit-developing feature of this application is helping people establishing the good habit that they want to develop, ensuring the task is finished in a long process.
2. In todo list interface, all checked boxes cannot be unchecked. This is to avoid users keep checking and unchecking the same task to get repeated rewards. To avoid a accidental false touch, a confirmation window will appear when ticking the checkboxes.
3. Only 1 bone is rewarded upon each completion of todo. This is because we do not expect users to spend too much time interacting with their digital pets, and hence avoid the objective of this application to be violated.

In MS2, we did not manage to complete the unit testing in our coding.

Interface testing/ unit testing are as follow,
content including:

1. testing correct user input, whether expected/actual output matches
2. testing abnormal user input, whether they will be successfully intercepted
3. testing margin: maximum/minimum input, empty input, numbers, unreadable messages,

Feature	Action	Expected perfomance	Actual performance
Todo - Entry	Enter normal test cases: eg. Mission1 Clicking "+", enter text	Task shown on screen, under "Todo"	✓
Todo - Checkbox	Tick a task created Clicking checkbox	Confirmation window pop up	✓
Todo - Checkbox,	Confirm completion	Checkbox checked	✓

Rewarding system	Clicking "confirm"	Bone +1	
LongTermGoal			
Entry	Enter normal test cases: eg. Run 30 km in a month, Quantity: 30 Clicking "+", enter text, confirm	LTG shown on screen, perfentage starts with 0% completion	✓
Submission	Enter respective daily portion Clicking paw icon, enter number, confirm	text box appear, percentage increase respectively after submission Paw icon turns grey, no further clicking allowed	✓
Deleting	Swipe left the target LTG, delete	Delete option shown upon swiping left, LTG deleted after clicking delete	✓
Rewarding system	Completing 1 LTG Test case quantity: 1, as submission function is locked after one submission a day for a particular LTG	10 bones added	✓
Woof Garden			
Standby animation	no action	same Gif of the sitting digital pet is repeated.	✓
Happiness Level	no action	Drop 20% in a day	
Interactive feature, Rewarding system, Happiness level	click Walking dog/Feeding/Bathing icon	Animation change, 1 Bone decreases, HL increases by 5%	✓ HL drop requires further testing
Bottom Directory	In Todo/LTG interface, Click LTG/Todo/WG directory	Go to respective interface in WG, no more bottom directory in Todo/LTG, bottom directory persist	✓
Top Directory	In WG, Click top left triangle	Back to Todo/LTG interface	✓
Abnormal test case Margin test case			

Todo			
Entry	Enter [blankspace]	Unable to add	✓
	Enter nothing	Unable to add	✓
	Enter 100 letters test case: 0123456789012345678901234567890123456789012345678901234567890123456789012345678901234567890123456789	Visible	✓
LTG			
Entry	Enter a LTG with quantity : 0		NaN% completed Change needed
	Enter a LTG with quantity : text	Cannot input text	✓
	Enter a LTG with quantity:[blankspace]	Cannot input blankspace	✓
	Enter a LTG with quantity: nothing	Cannot add	✓
	Enter a LTG with quantity: negetive number / Fraction	Cannot input negative number / fraction	✓
WG			
Rewarding system	When 0 bone, Clicking interaction icons	no bone window pop out no animation change	✓
Happiness Level	When 100%, have bones Clicking interaction icons	Respective bone deduction with No increase in HL	✓